

## ASSIGNMENT - 01

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Network Security.

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95/100 ✓

a) Encrypt "DEFEND THE WALL" with key = 3

Plain text:

DEFEND THE WALL

Cipher text:

GHIHROG WKH ZDOO

b) Decrypt "KHORR ZRUDG" with key = 3.

Cipher text:

KHORR ZRUDG

Plain text:

HELLO WORLD

c) Determine the key and decrypt "WKLVLVDWHW"

SECRET  $\rightarrow$  VHFUHW

The key is 3

Decrypted text:

THIS IS A TEST.

a) Encrypt "FIRE"

key: A  $\rightarrow$  Q, B  $\rightarrow$  W, C  $\rightarrow$  E, D  $\rightarrow$  R, ~~E  $\rightarrow$  T~~, F  $\rightarrow$  Y....

Encryption:

F  $\rightarrow$  Y, I  $\rightarrow$  O, R  $\rightarrow$  K, E  $\rightarrow$  T.

Cipher text:

YOKT.

2) Decrypt

Cipher text:

UTHHB QEB NRFZH YOLTK

Using frequency analysis, the best plaintext is:

HELLO THE QUICK BROWN

3) Vigenère Cipher

a) Encrypt "ATTACKATPALACE"

Key word: LEMON

Cipher text: LXFOPVEFRNHRMHA

b) Decrypt:

Cipher text: LXFOPVEFRNHR

Plain text: ATTACKATDAWN

c) Decrypt:

Cipher text: QNXEPKMAEGTU

Key word: KEY

Plain text: GIDTALAOQAWPA



Plaintext is

Hill Cipher  $\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

a) Encrypt "H3"

$$H = 7$$

$$3 = 8$$

Cipher text: TC

b) Plaintext: PO

$$P = 15$$

$$O = 14$$

Applying the matrix gives

QC.

Hence the encryption is verified.

5)

Playfair Cipher:

a) Encrypt:

Plain text:

MEET ME AFTER LUNCH

Keyword:

ZEBRAS

Cipher text:

UGIMR UGIPITDOVKAGL

b) Decrypt:  
Cipher text: EATTIANMTEFLUHE TREL  
Keyword: BOARD  
Plaintext: MEET ME AFTER

One Time Pad

a) Encrypt

Plain text: HELLO

Key : XMCKL

Cipher text: EQNVZ

b) Why OTP is unbreakable?

→ One-Time Pad is perfectly secure because

- The key is truly random.
- The key length equals the message length.
- The key is used only once.
- Without the key, every plain text is equally likely.

Hence OTP is theoretically impossible to crack.



## Columnar Transposition

a) Encrypt

Plaintext:

WE ARE DISCOVERED

Keyword: ZEBRA

Ciphertext: EVLNEAD TKESBAGROFOJDEECUXWN

b) Decrypt

EVLNEAD TKESBAGROFOJDEECUXWN

Keyword: TRIAL

Plaintext: WEAREDISCOVEREDLEETONCE

c) INFORMATION SECURITY

Keyword: HACK

- Plaintext length = 19
- Columns = 4
- Rows required = 5
- Padding required = 1 character(x)

Ciphertext:

(Arrange in Columns and read according to HACK key order)

8)

Frequency Analysis

Ciphertext:

VZQS DVUOHXMOFVUIPOZPEVSGNZWSZOFPPESXUZUE

Frequency analysis:  $\rightarrow$  Most frequent letters:

Z, V, O, P

These are likely to represent

E, T, A, O respectively.

Partial Decryption: Some words can be guessed through repeated patterns.

Best possible plaintext: THIS MESSAGE CONTAINS ENGLISH WORDS.

(The exact plaintext depends on the unknown substitution key.)

9)

Known Plaintext Attack

Ciphertext: BJJW AJW YM JW NX F XMRUAJWX

Known plaintext: THIS IS THE

Mappings obtained: B  $\rightarrow$  T, J  $\rightarrow$  H, W  $\rightarrow$  I, Q  $\rightarrow$  S, Y  $\rightarrow$  E

M  $\rightarrow$  T, N  $\rightarrow$  T, X  $\rightarrow$  S.

Complete plaintext: THIS IS THE A SIMPLEX.

cipher text: Z3CVTWANVRZGVTWAVZ4CAYHLMGTJ  
using frequency analysis and Kasiski examination,  
Recovered key: KEY

Decrypted plain text: WE ARE COVERED SAVE YOURSEL